Module 2:Transport Layer and Network Layer

Module 2:Transport Layer and Network Layer

Transport Layer

- Services and Protocols
- UDP and TCP

Hands-on

- Sockets Introduction
- Elementary TCP Sockets
- TCP Client/Server Example
- I/O Multiplexing
  - select()
  - poll()
- Elementary UDP Sockets
- Advanced I/O Functions

Network Layer

- Introduction
- Network-layer Protocols
- Unicast Routing
- Multicast Routing
- Multicasting Basics
- Intra-domain Routing
- Inter-domain Routing
- Next Generation IP (IPv6)
- Quality of Service (QoS)

Hands-on

- Linux Routing Table
- Routing Caches
- Routing Cache Implementation
- Adding Routes using `ip`

Multicast Routing

- Communication in the Internet is not limited to unicast communication.
- Many modern applications require one-to-many communication.
- Multicast Routing enables efficient delivery of packets from one sender to multiple receivers.
- Routers duplicate packets only when necessary.

Types of Communication

1. Unicast (One-to-One)
2. Multicast (One-to-Many)
3. Broadcast (One-to-All)

Unicasting

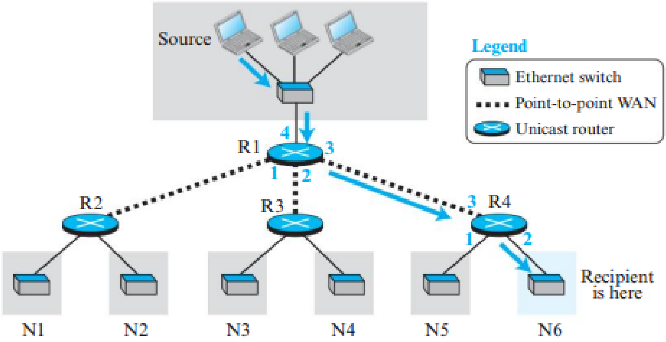
- One source communicates with one destination.
- Relationship: One-to-One.
- Destination address is a unicast IP address.
- Each router forwards the packet through only one interface.

$$\text{Source} \rightarrow \text{Router} \rightarrow \text{Router} \rightarrow \text{Destination}$$

Characteristics

- Most common communication model.
- Used in web browsing, file transfer, email retrieval, SSH, etc.

Figure 4.87 *Unicasting*



Multicasting

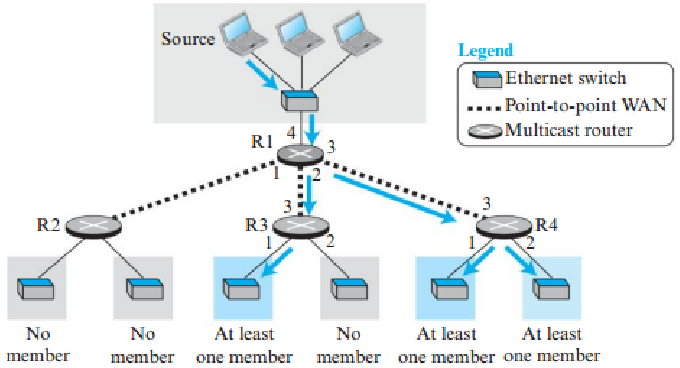
- One source sends data to multiple receivers.
- Relationship: One-to-Many.
- Destination address is a multicast group address.
- Routers replicate packets only when paths diverge.

$$\text{Source} \Rightarrow \text{Group of Receivers}$$

Characteristics

- Conserves bandwidth.
- Reduces sender workload.
- Efficient for large groups.

Figure 4.88 *Multicasting*



How Multicast Routers Work

- A multicast router maintains multicast forwarding information.
- If receivers exist in multiple directions, the router duplicates packets.
- Packets are forwarded only toward networks containing group members.

Example

- Router R1 forwards packets through Interfaces 2 and 3.
- Router R4 forwards through both interfaces.
- Router R3 forwards through only one interface because no group members exist in the other direction.

Advantage

Packet duplication occurs inside the network rather than at the source.

Multicasting vs Multiple Unicasting

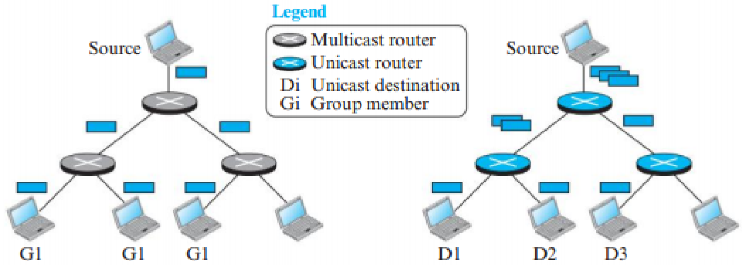
Multicasting

- Source sends only one packet.
- Routers duplicate packets.
- Same multicast destination address.
- Efficient bandwidth utilization.

Multiple Unicasting

- Source sends multiple packets.
- Each packet has a different destination.
- Duplicate traffic on many links.
- Higher bandwidth consumption.

Figure 4.89 *Multicasting versus multiple unicasting*



a. Multicasting

b. Multiple unicasting

Why Not Use Multiple Unicasting?

Reason 1: Bandwidth Efficiency

- Multicasting requires only one packet transmission from source.
- Duplicate packets are created only where necessary.
- Reduces network traffic significantly.

Reason 2: Reduced Delay

- In multiple unicasting, packets are created sequentially.
- Large recipient groups cause noticeable delay.
- Multicasting creates only one packet, eliminating source-side delays.

Advantages of Multicasting

1. Efficient use of bandwidth.
2. Reduced processing load at the sender.
3. Simultaneous delivery to many receivers.
4. Scalable communication model.
5. Lower transmission costs.
6. Suitable for real-time applications.

Key Benefit

One transmission from the source can reach thousands of receivers efficiently.

Multicast Applications

1. **Distributed Databases**
   - Query sent to multiple database locations simultaneously.
   - Database containing requested information responds.
2. **Information Dissemination**
   - Software updates.
   - News distribution.
   - Stock market updates.
3. **Live Media Streaming**
   - Video streaming.
   - Audio broadcasting.

More Applications of Multicasting

Teleconferencing

- Multiple participants receive identical content simultaneously.
- Used in online meetings and webinars.

Distance Learning

- One instructor teaches many students.
- Efficient content delivery.

Collaborative Applications

- Multiplayer gaming.
- Shared virtual environments.
- Group communication systems.

Broadcasting

- Broadcasting means One-to-All communication.
- A packet is delivered to every host in a network.
- Destination address is a broadcast address.

Examples

- DHCP Discover Messages
- ARP Requests
- Network Service Discovery

Limitation

Internet-wide broadcasting is not supported because it would generate enormous traffic and consume excessive bandwidth.

Comparison: Unicast vs Multicast vs Broadcast

Feature	Unicast	Multicast	Broadcast
Communication	One-to-One	One-to-Many	One-to-All
Destination	Single Host	Group	All Hosts
Bandwidth Usage	Medium	Low	High
Router Duplication	No	Yes	Yes
Scalability	Good	Excellent	Poor
Internet Support	Yes	Yes	Limited

Best Use Cases

- Unicast → Web browsing, Email
- Multicast → Streaming, Online classes
- Broadcast → DHCP, ARP

Summary

- Unicast provides one-to-one communication.
- Multicast provides efficient one-to-many communication.
- Broadcast provides one-to-all communication.
- Multicasting reduces bandwidth consumption and sender workload.
- Multicast routers duplicate packets only when required.
- Widely used in streaming, teleconferencing, distributed databases, and distance learning.

Key Takeaway

Multicast routing enables scalable and efficient group communication across networks.

Multicasting Basics

- Before implementing multicast routing, routers must understand:
  1. Multicast Addressing
  2. Group Membership Information
  3. Multicast Distribution Trees
- Multicasting requires additional mechanisms beyond unicast routing.

Key Challenge

Routers must determine which networks contain members of a multicast group.

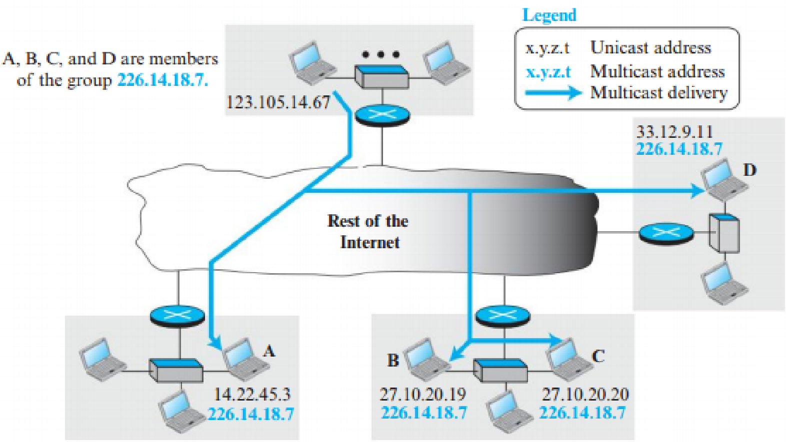
Multicast Addresses

- In unicast communication:
  - Source address identifies sender.
  - Destination address identifies a single receiver.
- In multicast communication:
  - Source address is a unicast address.
  - Destination address is a multicast group address.
- A multicast address identifies a group of receivers.

Important

A multicast address represents a group, not an individual host.

Figure 4.90 *Needs for multicast addresses*



Group Membership Concept

- A host may belong to multiple multicast groups.
- Each group has a unique multicast address.
- Hosts receive multicast packets only for groups they join.

Example

A host belonging to:

- Group G1
- Group G2
- Group G3

has:

- One Unicast Address
- Three Multicast Addresses

$$Total\ Addresses = n + 1$$

where $n =$ number of multicast groups.

IPv4 Multicast Address Range

- IPv4 reserves a special block for multicast communication.
- Classful Addressing:
  - Entire Class D
- Classless Addressing:
  - 224.0.0.0/4

$$224.0.0.0 \rightarrow 239.255.255.255$$

Size of Multicast Space

$$2^{28}$$

possible multicast addresses.

Figure 4.91 *A multicast address in binary*

Structure of IPv4 Multicast Addresses

- First four bits are fixed:

$$1110xxxxxxxxxxxxxxxxxxxxxxxxxxxx$$

- Prefix 1110 identifies multicast traffic.
- Remaining 28 bits identify multicast groups.

Observation

The multicast address identifies a group rather than a host.

Local Network Control Block

Address Range

$$224.0.0.0/24$$

- Used inside a local network.
- Packets are not forwarded by routers.

Important Addresses

- 224.0.0.0 $\rightarrow$ Reserved
- 224.0.0.1 $\rightarrow$ All hosts and routers
- 224.0.0.2 $\rightarrow$ All routers

Purpose

Supports multicast control and routing protocols within a LAN.

Internetwork Control Block

Address Range

$$224.0.1.0/24$$

- Used for Internet-wide multicast communication.
- Routers are allowed to forward these packets.

Use Case

Communication between multicast routers across different networks.

Source Specific Multicast (SSM)

Address Range

$$232.0.0.0/8$$

- Supports Source-Specific Multicast.
- Routing depends on:
  1. Source Address
  2. Group Address

Advantage

Allows unique identification using both source and multicast group.

GLOP Multicast Block

Address Range

$$233.0.0.0/8$$

- Used within an Autonomous System (AS).
- Uses the AS number as part of the multicast address.

Format

$$233.x.y.0 \rightarrow 233.x.y.255$$

where $x.y$ represents the AS number.

Benefit

Provides 256 multicast addresses per Autonomous System.

Administratively Scoped Block

Address Range

$$239.0.0.0/8$$

- Reserved for local organizational use.
- Packets should not leave the administrative domain.

Similar To

Private IPv4 addresses used in unicast communication.

Selecting a Multicast Address

The multicast address depends on the application.

Limited Group

- Uses AS-specific multicast addresses.
- Suitable for campus or enterprise networks.

Large Group

- Uses Source-Specific Multicast addresses.
- Suitable for Internet-wide communication.

Example: College Multicast Groups

Suppose a college belongs to:

$$AS = 23452$$

Convert AS number into two octets:

$$23452 = (91.156)_{256}$$

Available multicast addresses:

$$233.91.156.0$$

to

$$233.91.156.255$$

- Each professor may be assigned one multicast address.
- Students join the corresponding group.

Collecting Group Information

Before routing multicast packets, routers must know:

- Which multicast groups exist?
- Which interfaces contain members?

Challenge

Group membership changes dynamically.

Hosts may:

- Join groups
- Leave groups

at any time.

Unicast vs Multicast Advertisement

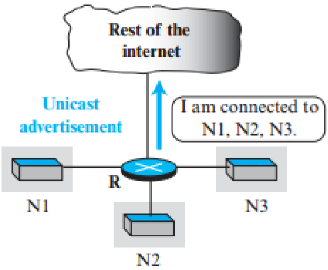
Unicast Routing

- Router automatically knows network prefixes.
- Prefixes are advertised using routing protocols.

Multicast Routing

- Router does not know group memberships automatically.
- Hosts must inform routers about group participation.

Figure 4.92 *Unicast versus multicast advertisement*



Why Extra Protocols Are Needed?

Unicast Routing

Requires:

- Routing Protocol Only

Multicast Routing

Requires:

1. Group Membership Protocol
2. Multicast Routing Protocol

Example

IGMP helps routers discover multicast group members.

Summary

- Multicast communication uses group addresses.
- IPv4 multicast range is 224.0.0.0/4.
- Several multicast subblocks serve different purposes.
- Hosts may belong to multiple multicast groups.
- Routers must learn group memberships dynamically.
- Multicast routing requires both membership management and routing protocols.

Key Takeaway

Multicast routing begins with identifying multicast groups and determining where their members exist.

Internet Group Management Protocol (IGMP)

- IGMP (Internet Group Management Protocol) is used to collect multicast group membership information.
- It operates at the Network Layer.
- Considered an auxiliary protocol of IP, similar to ICMP.
- IGMP messages are encapsulated inside IP datagrams.

Purpose

Allows routers to discover which multicast groups have active members on directly connected networks.

Why IGMP is Needed?

- Routers automatically know network prefixes in unicast routing.
- Routers do not automatically know multicast group memberships.
- Hosts may dynamically:
  - Join multicast groups
  - Leave multicast groups
- Routers require a mechanism to learn these changes.

Solution

IGMP provides communication between hosts and multicast routers.

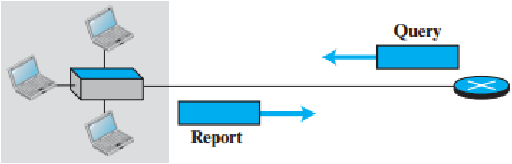
IGMPv3 Message Types

IGMP Version 3 uses only two message categories:

1. Query Messages
2. Report Messages

Router → Query
Host → Report

Figure 4.93 *IGMP operation*



IGMP Query Messages

- Query messages are sent periodically by multicast routers.
- Purpose:
    - Discover active multicast groups.
    - Verify existing memberships.

IGMPv3 supports three query types:

1. General Query
2. Group-Specific Query
3. Source-and-Group Specific Query

General Query Message

- Sent periodically to discover all multicast memberships.
- Destination Address:

$$224.0.0.1$$

(All Hosts and Routers)

- Every host receives the query.
- Routers hearing the query avoid sending duplicate queries.

Purpose

Ask all hosts: "Which multicast groups are you interested in?"

Group-Specific Query

- Used when a router wants information about a particular multicast group.
- Contains:
  - Multicast Group Address
- Encapsulated with destination equal to the multicast group address.

Example

Router asks:

"Are there any members of Group G1 still active?"

Hosts not belonging to the group simply ignore the message.

Source-and-Group Specific Query

- Most advanced IGMPv3 query type.
- Used when the router wants information about:
  - A specific multicast group
  - From a specific source

Example

"Are there hosts interested in Group G1 from Source S1?"

Benefit

Supports Source-Specific Multicast (SSM).

IGMP Report Messages

- Sent by hosts in response to router queries.
- Contains multicast membership information.
- Destination multicast address:

$$224.0.0.22$$

(Reserved for IGMPv3)

- Includes list of multicast groups.
- Includes preferred multicast sources.

Information in a Report Message

Each report contains records specifying:

1. Multicast Group Address
2. Included Sources
3. Excluded Sources

Inclusion Mode

Receive traffic only from:

$$\{S1, S2, S3\}$$

Exclusion Mode

Receive traffic from all except:

$$\{S4, S5\}$$

Joining and Leaving a Multicast Group

Joining a Group

1. Host receives query.
2. Host sends report.
3. Router records membership.

Leaving a Group

1. Host stops responding.
2. Router sends verification query.
3. If no response arrives, membership is removed.

IGMP Membership Process

Router Sends Query

$\Downarrow$

Hosts Respond with Reports

$\Downarrow$

Router Updates Membership Database

$\Downarrow$

Multicast Routing Protocol Uses Membership Information

Multicast Forwarding

Multicast forwarding differs from unicast forwarding.

Unicast

- One destination.
- Packet forwarded through one interface.

Multicast

- Multiple destinations.
- Router may forward through multiple interfaces.

Multicast Forwarding Example

Suppose multicast group G1 has members in different parts of the network. A router may need to:

- Send one copy through Interface 1.
- Send another copy through Interface 2.
- Send another copy through Interface 3.

Key Point

Routers replicate packets only when paths diverge.

Figure 4.94 *Destination in unicasting and multicasting*

a. Destination in unicasting is one

b. Destination in mulicasting is more than one

Forwarding Depends on Source and Destination

Unicast Routing

Forwarding decision depends only on:

$$Destination\ Address$$

Multicast Routing

Forwarding decision depends on:

$$(Source,\ Destination\ Group)$$

Reason

The router must know where the packet came from and where group members exist.

Why Source Information Matters?

Consider two situations:

1. Source located outside the multicast group region.
2. Source located inside a region containing group members.

The forwarding decision changes accordingly.

- Sending packets back toward already-served members creates duplicates.
- Routers avoid unnecessary packet replication.

Figure 4.95 *Forwarding depends on the destination and the source*

a. Packet sent out of two interfaces

b. Packet sent out of one interface

Two Approaches to Multicast Routing

Multicast routing uses two major approaches:

1. Source-Based Trees
2. Group-Shared Trees

Challenge

Multicast routing is more complex because both source and destination group influence routing.

Source-Based Tree Approach

- Separate multicast tree for each:

$$(Source,\ Group)$$

combination.

If:

- m = number of groups
- n = number of sources

Then total trees:

$$m \times n$$

Characteristics

- Highly optimized paths.
- Larger storage requirements.
- More routing complexity.

Source-Based Trees Example

Suppose:

- 10 Groups
- 100 Sources

Number of multicast trees:

$$10 \times 100 = 1000$$

Each tree:

- Root = Source
- Leaves = Group Members

Advantage

Shortest path from source to receivers.

Group-Shared Tree Approach

- One router acts as a central multicast point.
- Called:
  - Core Router
  - Rendezvous Point (RP)

All multicast sources send traffic to the RP.
The RP distributes packets to group members.

Idea

One shared tree per group.

Group-Shared Tree Architecture

Source

$\Downarrow$

Rendezvous Point (Core Router)

$\Downarrow$

Multicast Group Members

The RP becomes the root of the multicast tree.

Tree Count Comparison

Suppose:

- m Groups
- n Sources

Approach	Number of Trees
Source-Based	$m \times n$
Group-Shared	m

Observation

Group-shared trees significantly reduce routing state information.

Tunneling in Group-Shared Trees

The source first sends the multicast packet to the RP.
This requires:

- Encapsulation
- Tunneling

Process:

1. Multicast packet encapsulated in unicast packet.
2. Sent to RP.
3. RP decapsulates packet.
4. RP multicasts to receivers.

Source-Based vs Group-Shared Trees

Feature	Source-Based Tree	Group-Shared Tree
Root Node	Source	RP/Core Router
Trees Required	$m \times n$	m
Storage Requirement	High	Low
Path Efficiency	Excellent	Moderate
Complexity	High	Lower

Summary

- IGMP collects multicast membership information.
- IGMPv3 uses Query and Report messages.
- Routers periodically query hosts for membership information.
- Multicast forwarding depends on both source and destination group.
- Source-Based Trees create one tree per source-group pair.
- Group-Shared Trees use a Rendezvous Point (RP) to reduce routing complexity.

Key Takeaway

IGMP enables multicast group management, while multicast routing protocols use this information to build efficient multicast delivery trees.

Module 2:Transport Layer and Network Layer

Module 2:Transport Layer and Network Layer

Transport Layer

- Services and Protocols
- UDP and TCP

Hands-on

- Sockets Introduction
- Elementary TCP Sockets
- TCP Client/Server Example
- I/O Multiplexing
  - select()
  - poll()
- Elementary UDP Sockets
- Advanced I/O Functions

Network Layer

- Introduction
- Network-layer Protocols
- Unicast Routing
- Multicast Routing
- Multicasting Basics
- Intra-domain Routing
- Inter-domain Routing
- Next Generation IP (IPv6)
- Quality of Service (QoS)

Hands-on

- Linux Routing Table
- Routing Caches
- Routing Cache Implementation
- Adding Routes using `ip`

Intradomain Multicast Routing Protocols

- Several multicast routing protocols have been developed.
- These protocols operate within an Autonomous System (AS).
- Two protocols extend existing unicast routing protocols.
- One protocol is independent and widely used.

Protocols Covered

1. DVMRP (Distance Vector Multicast Routing Protocol)
2. MOSPF (Multicast Open Shortest Path First)
3. PIM (Protocol Independent Multicast) *(next section)*

Source-Based Tree Approach

Both DVMRP and MOSPF use the:

Source-Based Tree Approach

- One multicast tree for every Source-Group pair.
- Source becomes the root of the tree.
- Group members become leaves.

Advantage

Provides shortest path delivery from source to receivers.

Drawback

Large number of multicast trees must be maintained.

Distance Vector Multicast Routing Protocol (DVMRP)

- Extension of RIP for multicast routing.
- Uses Source-Based Trees.
- Builds multicast trees dynamically.
- Uses three algorithms:

1. Reverse Path Forwarding (RPF)
2. Reverse Path Broadcasting (RPB)
3. Reverse Path Multicasting (RPM)

Goal

Deliver multicast packets only to interested receivers while preventing loops.

DVMRP Operation Overview

Source Packet

$\Downarrow$

Reverse Path Forwarding (RPF)

$\Downarrow$

Reverse Path Broadcasting (RPB)

$\Downarrow$

Reverse Path Multicasting (RPM)

$\Downarrow$

Multicast Delivery Tree

Reverse Path Forwarding (RPF)

RPF ensures that routers accept multicast packets only from the correct direction.

Rule

A multicast packet is accepted only if it arrives through the interface that lies on the shortest path back to the source.

Router consults its unicast routing table:

- Pretend destination = Source
- Find outgoing interface
- Accept multicast packet only from that interface

Why RPF is Needed?

Without RPF:

- Routers may receive duplicate multicast packets.
- Packets can circulate indefinitely.
- Routing loops may occur.

With RPF:

- Only one valid copy is accepted.
- Duplicate packets are discarded.
- Loops are prevented.

RPF Example

Assume:

- Source = S
- Router = R

Routing table indicates:

$$R \rightarrow Source$$

should use Interface 2.

Then:

- Packet arriving on Interface 2 $\rightarrow$ Accepted
- Packet arriving on Interface 1 or 3 $\rightarrow$ Dropped

Result

Only shortest-path packets survive.

Reverse Path Broadcasting (RPB)

RPF prevents router loops.
However:

- Networks may still receive duplicate copies.
- Multiple routers connected to same LAN may forward identical packets.

Solution

RPB selects a designated parent router for each network.

Parent Router Selection

Each network chooses one parent router.
Selection criteria:

1. Router with shortest path to source.
2. If tie occurs:
    - Router with smaller IP address wins.

Only the parent router forwards multicast packets onto the network.

RPB Operation

- Parent router forwards multicast packet.
- Non-parent routers suppress forwarding.
- Duplicate packet delivery is avoided.

Outcome

A shortest-path broadcast tree is formed.

Root:

$$Source$$

Leaves:

$$All\ Networks$$

Insert Figure 4.96

Result of RPB

After Reverse Path Broadcasting:

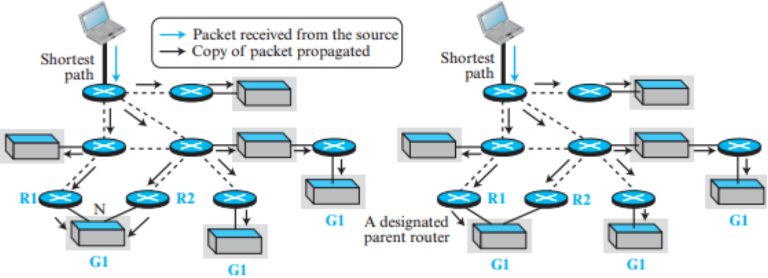
- Every network receives exactly one copy.
- No duplicate LAN delivery.
- Shortest path maintained.

However:

Problem

Packets are still broadcast to all networks, including those with no multicast members.

Figure 4.96 *RPF versus RPB*



a. Using RPF, N receives two copies.

b. Using RPB, N receives only one copy.

Reverse Path Multicasting (RPM)

RPM converts the broadcast tree into a multicast tree.

Idea

Prune branches that do not contain multicast group members.

Result:

- Only interested networks receive packets.
- Bandwidth usage decreases.
- Efficiency improves.

RPM Pruning Process

1. Routers collect membership information using IGMP.
2. Leaf routers determine active groups.
3. Membership information propagates upward.
4. Routers prune unnecessary branches.

Effect

Broadcast Tree $\rightarrow$ Multicast Tree

RPM Example

Networks:

$$N_1, N_2, N_3, N_4, N_5$$

Suppose only:

$$N_2 \text{ and } N_5$$

contain group members.

RPM removes branches leading to:

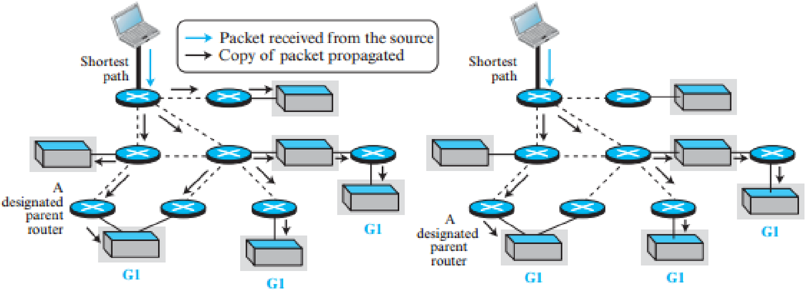
$$N_1, N_3, N_4$$

Thus packets travel only where needed.

DVMRP Summary

Algorithm	Function
RPF	Prevent loops
RPB	Build broadcast tree
RPM	Prune tree to multicast tree

Figure 4.97 *RPB versus RPM*



a. Using RPB, all networks receive a copy.

b. Using RPM, only members receive a copy.

Multicast Open Shortest Path First (MOSPF)

- Extension of OSPF for multicast routing.
- Uses Source-Based Trees.
- Based on Link-State Routing.

Key Difference

MOSPF uses Link-State Database (LSDB) instead of distance vectors.

MOSPF Architecture

Each router maintains:

1. Standard OSPF Link-State Database (LSDB)
2. Multicast Group Membership Database

The LSDB contains:

- Complete network topology.
- Link costs.

Membership database contains:

- Active multicast groups.

MOSPF Multicast Forwarding Process

When router receives packet:

$$(Source = S,\ Group = G)$$

it performs:

1. Build shortest-path tree rooted at S.
2. Locate itself in that tree.
3. Create broadcast subtree.
4. Prune non-member branches.

Step 1: Build Source-Based Tree

Router uses:

Dijkstra's Algorithm

on LSDB.

Tree characteristics:

- Root = Source S
- Leaves = All Networks

Important

This tree differs from ordinary OSPF routing where the router itself is the root.

Step 2: Create Local Subtree

After constructing the source tree:

- Router finds its own position.
- Creates subtree rooted at itself.

The subtree contains all downstream destinations.

Purpose

Determine which interfaces may require packet forwarding.

Step 3: Prune the Broadcast Tree

MOSPF initially creates:

$$Broadcast\ Tree$$

using shortest paths.

Then:

- Membership information collected using IGMP.
- Routers flood membership updates.
- Branches without members are removed.

Result:

$$Multicast\ Tree$$

Step 4: Forward Packet

After pruning:

- Router forwards packet only through multicast branches.
- Interfaces without receivers are ignored.

Goal

Reach every multicast member exactly once.

MOSPF Example

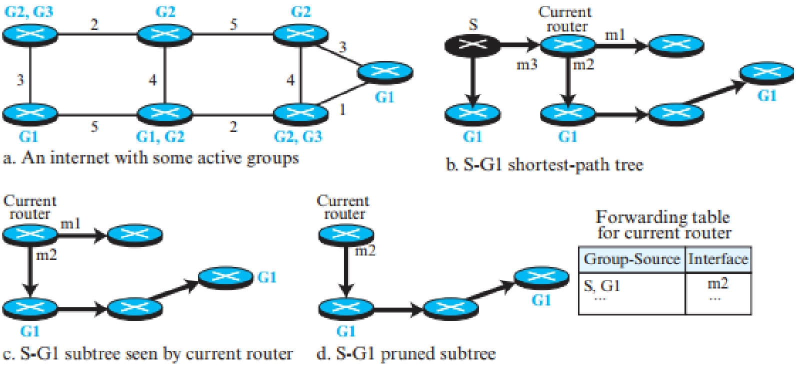
Router receives:

$$(Source = S,\ Group = G)$$

Process:

1. Create shortest-path tree rooted at S.
2. Extract local subtree.
3. Remove branches without G members.
4. Forward packet.

Figure 4.98 *Example of tree formation in MOSPF*



DVMRP vs MOSPF

Feature	DVMRP	MOSPF
Based On	RIP	OSPF
Routing Method	Distance Vector	Link State
Tree Type	Source-Based	Source-Based
Loop Prevention	RPF	LSDB + Dijkstra
Membership Info	IGMP + Pruning	IGMP + LS Updates
Scalability	Moderate	Better
Computation	Lower	Higher

Advantages and Limitations

DVMRP

- Simple implementation.
- Efficient pruning.
- Higher overhead due to broadcast-first approach.

MOSPF

- Uses complete network topology.
- More accurate multicast trees.
- Requires more computation and memory.

Summary

- DVMRP extends RIP for multicast routing.
- Uses RPF, RPB and RPM algorithms.
- MOSPF extends OSPF for multicast routing.
- Uses LSDB and Dijkstra's algorithm.
- Both protocols use Source-Based Trees.
- Membership information is collected through IGMP.

Key Takeaway

DVMRP builds multicast trees through reverse-path techniques, while MOSPF builds multicast trees using shortest-path computations from complete topology information.

Protocol Independent Multicast (PIM)

- PIM is one of the most widely used multicast routing protocols.
- It is called **Protocol Independent** because it can work with:
  - Distance Vector routing protocols
  - Link State routing protocols
- PIM uses the forwarding table created by the underlying unicast routing protocol.
- It does not care how the forwarding table is built.

Key Idea

PIM relies on existing unicast routing information for multicast forwarding.

Operating Modes of PIM

PIM operates in two modes:

1. **PIM-DM (Dense Mode)**
2. **PIM-SM (Sparse Mode)**

Dense Mode

- Large number of multicast members.
- High probability that routers have group members.

Sparse Mode

- Small number of multicast members.
- Low probability that routers have group members.

Dense vs Sparse Groups

Dense Group	Sparse Group
Many active members	Few active members
Routers likely have members	Routers rarely have members
Broadcast then prune	Build tree first
Uses Source-Based Trees	Uses Shared Trees
PIM-DM	PIM-SM

Examples

- Dense: Popular live sports streaming
- Sparse: Specialized research webinar

PIM Dense Mode (PIM-DM)

- Used when multicast group members are densely distributed.
- Uses the Source-Based Tree approach.
- Similar to DVMRP but much simpler.
- Uses:
  1. Reverse Path Forwarding (RPF)
  2. Reverse Path Multicasting (RPM)
- Does not wait for pruning before forwarding packets.

PIM-DM Working Principle

Source Packet

$\Downarrow$

RPF Check

$\Downarrow$

Broadcast Packet

$\Downarrow$

Routers without members send Prune Messages

$\Downarrow$

Broadcast Tree becomes Multicast Tree

Step 1: Reverse Path Forwarding (RPF)

When a router receives multicast packet:

$$(Source = S,\ Group = G)$$

the router:

1. Checks unicast forwarding table.
2. Finds shortest reverse path to source S.
3. Accepts packet only if it arrives through that interface.

If Packet Arrives Through Wrong Interface

Router drops packet and sends a prune message.

RPF Example

Assume:

- Router R reaches Source S through Interface 2.

Then:

- Packet arriving on Interface 2 $\rightarrow$ Accepted
- Packet arriving on Interface 1 $\rightarrow$ Dropped
- Packet arriving on Interface 3 $\rightarrow$ Dropped

Benefit

Prevents multicast loops and duplicate packets.

Step 2: Multicast Forwarding

If packet passes RPF test:

- Router forwards packet through all interfaces
- Except:
  - Incoming interface
  - Interfaces already pruned

Initially:

$$Broadcasting$$

occurs.

Later:

$$Pruning$$

converts broadcast tree into multicast tree.

Prune Messages in PIM-DM

Routers without interested members:

- Send prune messages upstream.
- Tell neighboring routers to stop forwarding packets.

Gradually:

Broadcast Tree

becomes

Efficient Multicast Tree

DVMRP vs PIM-DM

Feature	DVMRP	PIM-DM
Uses RPF	Yes	Yes
Uses RPM	Yes	Yes
Uses RPB	Yes	No
Waits for Pruning	Yes	No
Initial Behavior	Controlled	Flood First
Complexity	Higher	Lower

Advantages of PIM-DM

- Simple protocol.
- Fast multicast delivery.
- Effective for dense multicast groups.
- Requires minimal multicast state information.

Best Use Case

Large-scale live streaming where most networks contain receivers.

Limitations of PIM-DM

- Initial flooding wastes bandwidth.
- Not suitable for sparse groups.
- Unnecessary traffic reaches routers without members.

Problem

Many packets are forwarded before pruning occurs.

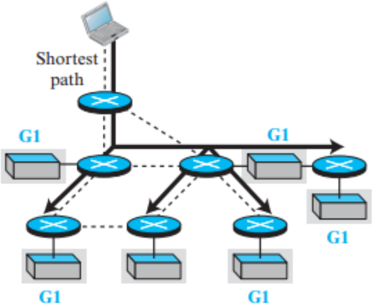
PIM-DM Example

- First multicast packet is broadcast everywhere.
- Routers without members send prune messages.
- Subsequent packets follow only multicast branches.

Result

Efficient multicast delivery after pruning.

Figure 4.99 *Idea behind PIM-DM*



a. First packet is broadcast.

b. Second packet is multicast.

Protocol Independent Multicast-Sparse Mode (PIM-SM)

- Designed for sparse multicast groups.
- Uses Group-Shared Tree approach.
- Avoids flooding.
- More scalable than PIM-DM.

Key Idea

Only routers interested in the group participate in multicast delivery.

Rendezvous Point (RP)

PIM-SM introduces a special router:

Rendezvous Point (RP)

Functions:

- Acts as multicast meeting point.
- Receives packets from sources.
- Distributes packets to group members.

Core Router

RP is equivalent to the Core Router in shared-tree multicasting.

Multicast Delivery in PIM-SM

Communication occurs in two stages.

Stage 1
Source → RP

Stage 2
RP → Multicast Group Members

This eliminates unnecessary flooding.

Tunneling to the RP

When source wants to send multicast data:

1. Multicast packet is encapsulated.
2. Encapsulated packet becomes unicast.
3. Sent to RP.
4. RP decapsulates packet.
5. RP multicasts packet to receivers.

Technique Used

Tunneling

Why Use an RP?

Without RP:

- Every router needs source-based trees.
- Number of trees = m $\times$ n

where

- m = groups
- n = sources

With RP:

$$m \text{ trees only}$$

Benefit

Significant reduction in routing state.

Multicast Tree in PIM-SM

Tree Structure:

- Root = Rendezvous Point (RP)
- Leaves = Designated Routers

Every designated router:

- Represents a network containing active members.

Join Messages

When a router discovers a group member using IGMP:

1. Router sends Join message.
2. Message travels toward RP.
3. Each router updates multicast forwarding table.

Purpose

Add a new branch to multicast tree.

How Join Messages Build the Tree

As Join message travels:

- Incoming interface marked as outgoing multicast interface.
- Outgoing interface toward RP marked as incoming multicast interface.

Gradually:

$$RP$$

becomes connected to all active members.

Prune Messages

When no active members remain:

1. Router sends Prune message.
2. Message travels toward RP.
3. Join counter is decremented.

Purpose

Remove unnecessary multicast branches.

Pruning Operation

If Join Count becomes:

$$0$$

then:

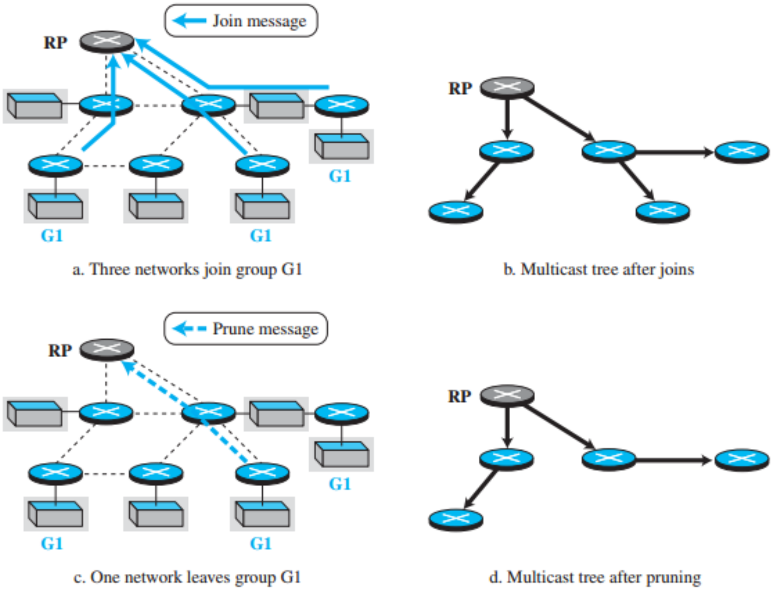
- Interface is removed from multicast tree.
- No multicast packets forwarded there.

This keeps multicast trees efficient.

Join and Prune Example

- Three networks join Group G1.
- Shared multicast tree is created.
- One network leaves.
- Corresponding branch is pruned.

Figure 4.100 *Idea behind PIM-SM*



PIM-DM vs PIM-SM

Feature	PIM-DM	PIM-SM
Group Density	High	Low
Tree Type	Source-Based	Shared Tree
Flooding	Yes	No
Core Router	Not Required	RP Required
Scalability	Moderate	High
Bandwidth Usage	Higher	Lower
Complexity	Lower	Higher

Advantages of PIM-SM

- Efficient bandwidth utilization.
- No unnecessary flooding.
- Suitable for Internet-scale multicast.
- Scales well for sparse groups.

Best Use Case

Large distributed multicast applications with relatively few receivers.

Limitations of PIM-SM

- RP selection is complex.
- Additional overhead due to tunneling.
- Failure of RP can affect multicast delivery.

Summary of PIM

- PIM is independent of underlying unicast routing protocol.
- Uses existing routing tables for multicast forwarding.
- Supports:
  - PIM-DM (Dense Mode)
  - PIM-SM (Sparse Mode)
- PIM-DM uses flooding and pruning.
- PIM-SM uses Rendezvous Point and shared trees.

Key Takeaway

PIM-DM is suitable for dense receiver distribution, while PIM-SM is the preferred solution for large sparse multicast environments.

Interdomain Multicast Routing

- DVMRP, MOSPF, and PIM operate **within an Autonomous System (AS)**.
- When multicast group members are distributed across multiple ASs, an **Interdomain Multicast Routing Protocol** is required.
- Interdomain multicast routing enables multicast communication across the global Internet.

Challenge

How can multicast packets efficiently travel between different Autonomous Systems?

Need for Interdomain Multicast Routing

Consider:

- Source located in AS1
- Some receivers located in AS2
- Others located in AS3

Intradomain protocols cannot:

- Exchange multicast information between ASs
- Build multicast trees across AS boundaries

Solution:

Interdomain Multicast Routing Protocols

Interdomain Multicast Protocols

Common protocols include:

1. Multicast Border Gateway Protocol (MBGP)
2. Multicast Source Discovery Protocol (MSDP)
3. Border Gateway Multicast Protocol (BGMP)

Goal

Provide multicast communication among multiple Autonomous Systems.

Multicast Border Gateway Protocol (MBGP)

- Extension of BGP used in unicast routing.
- Supports multicast routing between Autonomous Systems.
- Exchanges multicast routing information among border routers.

Main Idea

Maintain separate routing information for:

1. Unicast traffic
2. Multicast traffic

Dual Routing Paths in MBGP

MBGP maintains two independent paths:

Unicast Path

$$Source \rightarrow Destination$$

used for ordinary communication.

Multicast Path

$$Source \rightarrow Group$$

used for multicast communication.

This allows multicast traffic to follow routes different from unicast traffic.

MBGP Architecture

AS1 AS2 AS3

Border Routers exchange:

- Reachability information
- Multicast group information

Result

Each AS knows how to forward multicast traffic toward destination groups.

Shared-Group Tree in MBGP

MBGP uses:

Shared-Group Multicast Routing

Each Autonomous System selects:

Rendezvous Point (RP)

Responsibilities:

- Receive multicast traffic
- Distribute traffic inside the AS
- Cooperate with RPs in other ASs

Advantages of MBGP

- Reuses existing BGP infrastructure.
- Supports multicast communication across AS boundaries.
- Allows separate optimization of multicast routes.
- Scalable for large networks.

Limitations of MBGP

- Difficult for Rendezvous Points (RPs) to discover multicast sources located in other ASs.
- Additional mechanisms are required to advertise source information.

Problem

How does an RP learn about multicast sources in remote Autonomous Systems?

Multicast Source Discovery Protocol (MSDP)

- Proposed to overcome MBGP limitations.
- Works together with MBGP.
- Helps Rendezvous Points discover multicast sources located in other ASs.

Purpose

Provide source discovery across Autonomous Systems.

Why MSDP is Needed

Without MSDP:

- RP knows local sources.
- RP cannot easily discover remote multicast sources.

With MSDP:

- Source information is exchanged among ASs.
- RPs learn about multicast sources globally.

MSDP Architecture

Each Autonomous System designates:

Source Representative Router

Functions:

- Detect multicast sources within its AS.
- Advertise source information to RPs in other ASs.

Result

All RPs become aware of active multicast sources.

Operation of MSDP

1. Source begins multicast transmission.
2. Source Representative Router detects source.
3. MSDP advertises source information.
4. Remote RPs receive advertisement.
5. Interested receivers join multicast session.

Benefits of MSDP

- Solves source discovery problem.
- Enables global multicast communication.
- Works together with MBGP and PIM-SM.

Key Benefit

Rendezvous Points can locate multicast sources in remote Autonomous Systems.

Border Gateway Multicast Protocol (BGMP)

- Proposed as a possible replacement for MBGP.
- Designed specifically for interdomain multicast routing.
- Uses a shared multicast tree spanning multiple ASs.

Shared Tree Concept in BGMP

Unlike MBGP:

- Only one shared multicast tree exists for each group.
- Tree spans multiple Autonomous Systems.

Tree Components:

- Root located in one AS
- Leaves located in multiple ASs

BGMP Multicast Architecture

Root AS

$\Downarrow$

Shared Multicast Tree

$\Downarrow$

Multiple Autonomous Systems

All multicast traffic flows through the shared interdomain tree.

Advantages of BGMP

- Single multicast tree per group.
- Reduced routing state information.
- More efficient interdomain multicast management.

Challenges in BGMP

Two major problems must be solved:

1. Selecting the Autonomous System that becomes the root.
2. Informing all multicast sources about the root location.

Sources must know where to tunnel multicast packets.

Comparison of Interdomain Multicast Protocols

Protocol	Main Function	Key Feature
MBGP	Multicast routing between ASs	Separate multicast paths
MSDP	Source discovery	Advertises multicast sources
BGMP	Shared multicast tree	Single tree per group

MBGP vs MSDP vs BGMP

Feature	MBGP	MSDP	BGMP
Interdomain Routing	Yes	No	Yes
Source Discovery	No	Yes	Yes
Shared Tree	Yes	No	Yes
Based on BGP	Yes	No	Related
Primary Goal	Routing	Source Discovery	Global Tree

Summary

- Interdomain multicast routing is required when multicast members exist in multiple ASs.
- MBGP extends BGP to support multicast routing.
- MSDP helps Rendezvous Points discover multicast sources.
- BGMP proposes a single shared multicast tree spanning multiple ASs.

Key Takeaway

MBGP provides multicast connectivity, MSDP provides source discovery, and BGMP provides a scalable shared-tree solution for global multicast communication.